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第 25 章 Angular momentum operator

§25.1 Angular momentum operator

$$\mathbf{L} = \mathbf{r} \times \mathbf{p} = \mathbf{r} \times (-i\hbar\nabla) = -i\hbar\mathbf{r} \times \nabla. \quad (25.1.1)$$

$$l_x = yp_z - zp_y = -i\hbar\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right), \quad (25.1.2)$$

$$l_y = zp_x - xp_z = -i\hbar\left(z\frac{\partial}{\partial x} - x\frac{\partial}{\partial z}\right), \quad (25.1.3)$$

$$l_z = xp_y - yp_x = -i\hbar\left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}\right). \quad (25.1.4)$$

$$\mathbf{L}^2 := l_x^2 + l_y^2 + l_z^2. \quad (25.1.5)$$

对易子 (comutator):

$$[a, b] := ab - ba. \quad (25.1.6)$$

$$[a + b, c] = [a, c] + [b, c], \quad [a, b + c] = [a, b] + [a, c]. \quad (25.1.7)$$

$$[\lambda a, b] = \lambda[a, b], \quad [a, \lambda b] = \lambda[a, b]. \quad (25.1.8)$$

$$[ab, c] = a[b, c] + [a, c]b, \quad [a, bc] = [a, b]c + b[a, c]. \quad (25.1.9)$$

$$x_1 = x, \quad x_2 = y, \quad x_3 = z. \quad (25.1.10)$$

$$p_1 = p_x, \quad p_2 = p_y, \quad p_3 = p_z. \quad (25.1.11)$$

$$[x_i, x_j] = 0, \quad [p_i, p_j] = 0, \quad [x_i, p_j] = i\hbar\delta_{ij}. \quad (25.1.12)$$

$$[l_x, l_y] = i\hbar l_z, \quad [l_y, l_z] = i\hbar l_x, \quad [l_z, l_x] = i\hbar l_y. \quad (25.1.13)$$

证明.

$$\begin{aligned}
[l_x, l_y] &= [yp_z - zp_y, zp_x - xp_z] \\
&= [yp_z, zp_x - xp_z] - [zp_y, zp_x - xp_z] \\
&= [yp_z, zp_x] - [yp_z, xp_z] - [zp_y, zp_x] + [zp_y, xp_z] \\
&= y[p_z, zp_x] + [y, zp_x]p_z - [yp_z, xp_z] - [zp_y, zp_x] + [zp_y, xp_z] \\
&= y[p_z, z]p_x + yz[p_z, p_x] + [y, zp_x]p_z - [yp_z, xp_z] - [zp_y, zp_x] + [zp_y, xp_z] \\
&= -i\hbar yp_x + i\hbar xp_y \\
&= i\hbar(xp_y - yp_x) \\
&= i\hbar l_z.
\end{aligned} \quad (25.1.14)$$

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$$[l_x, \mathbf{l}^2] = 0, \quad [l_y, \mathbf{l}^2] = 0, \quad [l_z, \mathbf{l}^2] = 0. \quad (25.1.15)$$

证明.

$$\begin{aligned}
[l_x, \mathbf{l}^2] &= [l_x, l_x^2 + l_y^2 + l_z^2] \\
&= [l_x, l_x^2] + [l_x, l_y^2] + [l_x, l_z^2] \\
&= [l_x, l_y^2] + [l_x, l_z^2] \\
&= l_y[l_x, l_y] + [l_x, l_y]l_y + l_z[l_x, l_z] + [l_x, l_z]l_z \\
&= i\hbar l_y l_z + i\hbar l_z l_y - i\hbar l_z l_y - i\hbar l_y l_z \\
&= 0.
\end{aligned} \quad (25.1.16)$$

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§25.2 In spherical coordinate system

$$\nabla\psi = \mathbf{e}_r \frac{\partial\psi}{\partial r} + \mathbf{e}_\theta \frac{1}{r} \frac{\partial\psi}{\partial\theta} + \mathbf{e}_\varphi \frac{1}{r \sin\theta} \frac{\partial\psi}{\partial\varphi}. \quad (25.2.1)$$

$$\nabla = \mathbf{e}_r \frac{\partial}{\partial r} + \mathbf{e}_\theta \frac{1}{r} \frac{\partial}{\partial\theta} + \mathbf{e}_\varphi \frac{1}{r \sin\theta} \frac{\partial}{\partial\varphi}. \quad (25.2.2)$$

$$\mathbf{r} = r\mathbf{e}_r. \quad (25.2.3)$$

$$\begin{aligned}
\mathbf{L} &= -i\hbar \mathbf{r} \times \nabla \\
&= -i\hbar(r\mathbf{e}_r) \times \left(\mathbf{e}_r \frac{\partial}{\partial r} + \mathbf{e}_\theta \frac{1}{r} \frac{\partial}{\partial \theta} + \mathbf{e}_\varphi \frac{1}{r \sin \theta} \frac{\partial}{\partial \varphi} \right) \\
&= i\hbar \left(\mathbf{e}_\theta \frac{1}{\sin \theta} \frac{\partial}{\partial \varphi} - \mathbf{e}_\varphi \frac{\partial}{\partial \theta} \right). \tag{25.2.4}
\end{aligned}$$

$$\begin{aligned}
\mathbf{L}^2 &= \mathbf{L} \cdot \mathbf{L} \\
&= -\hbar^2 \left(\mathbf{e}_\theta \frac{1}{\sin \theta} \frac{\partial}{\partial \varphi} - \mathbf{e}_\varphi \frac{\partial}{\partial \theta} \right) \cdot \left(\mathbf{e}_\theta \frac{1}{\sin \theta} \frac{\partial}{\partial \varphi} - \mathbf{e}_\varphi \frac{\partial}{\partial \theta} \right) \\
&= -\hbar^2 \left(\frac{1}{\sin^2 \theta} \mathbf{e}_\theta \cdot \frac{\partial \mathbf{e}_\theta}{\partial \varphi} \frac{\partial}{\partial \varphi} + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \varphi^2} - \frac{1}{\sin \theta} \mathbf{e}_\theta \cdot \frac{\partial \mathbf{e}_\varphi}{\partial \theta} \frac{\partial}{\partial \theta} \right. \\
&\quad \left. - \frac{1}{\sin \theta} \mathbf{e}_\varphi \cdot \frac{\partial \mathbf{e}_\theta}{\partial \theta} \frac{\partial}{\partial \varphi} + \mathbf{e}_\varphi \cdot \frac{\partial \mathbf{e}_\varphi}{\partial \theta} \frac{\partial}{\partial \theta} + \frac{\partial^2}{\partial \theta^2} \right). \tag{25.2.5}
\end{aligned}$$

$$\mathbf{e}_r = \sin \theta \cos \varphi \mathbf{i} + \sin \theta \sin \varphi \mathbf{j} + \cos \theta \mathbf{k}, \tag{25.2.6}$$

$$\mathbf{e}_\theta = \cos \theta \cos \varphi \mathbf{i} + \cos \theta \sin \varphi \mathbf{j} - \sin \theta \mathbf{k}, \tag{25.2.7}$$

$$\mathbf{e}_\varphi = -\sin \varphi \mathbf{i} + \cos \varphi \mathbf{j}. \tag{25.2.8}$$

$$\mathbf{e}_\theta \cdot \frac{\partial \mathbf{e}_\theta}{\partial \varphi} = 0, \tag{25.2.9}$$

$$\mathbf{e}_\theta \cdot \frac{\partial \mathbf{e}_\varphi}{\partial \varphi} = -\cos \theta, \tag{25.2.10}$$

$$\mathbf{e}_\varphi \cdot \frac{\partial \mathbf{e}_\theta}{\partial \theta} = 0, \tag{25.2.11}$$

$$\mathbf{e}_\varphi \cdot \frac{\partial \mathbf{e}_\varphi}{\partial \theta} = 0. \tag{25.2.12}$$

$$\begin{aligned}
\mathbf{L}^2 &= -\hbar^2 \left(\frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \varphi^2} + \frac{\cos \theta}{\sin \theta} \frac{\partial}{\partial \theta} + \frac{\partial^2}{\partial \theta^2} \right) \\
&= -\hbar^2 \left(\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \sin \theta \frac{\partial}{\partial \theta} + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \varphi^2} \right). \tag{25.2.13}
\end{aligned}$$

$$\begin{aligned}
l_z &= \mathbf{k} \cdot \mathbf{L} \\
&= i\hbar \mathbf{k} \cdot \left(\mathbf{e}_\theta \frac{1}{\sin \theta} \frac{\partial}{\partial \varphi} - \mathbf{e}_\varphi \frac{\partial}{\partial \theta} \right) \\
&= i\hbar \left(\mathbf{k} \cdot \mathbf{e}_\theta \frac{1}{\sin \theta} \frac{\partial}{\partial \varphi} - \mathbf{k} \cdot \mathbf{e}_\varphi \frac{\partial}{\partial \theta} \right) \\
&= -i\hbar \frac{\partial}{\partial \varphi}. \tag{25.2.14}
\end{aligned}$$

$$l_z^2 = -\hbar^2 \frac{\partial^2}{\partial \varphi^2}. \quad (25.2.15)$$

$$\mathbf{L}^2 = -\hbar^2 \left(\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \sin \theta \frac{\partial}{\partial \theta} + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \varphi^2} \right). \quad (25.2.16)$$

$$l_z = -i\hbar \frac{\partial}{\partial \varphi}. \quad (25.2.17)$$